Hints: Differentiable functions – 2

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The expression $\xi f'(\xi) - f(\xi)$ is the numerator of the derivative of $f(x)/x$. Use the multiplier $\frac{1}{x}$, adjusted so that the function vanishes at a and b.

Hint 2. Apply Rolle to $g(x) = \dfrac{f(x) - k}{x}$ on $[a, b]$: $g'(x) = \dfrac{xf'(x) - f(x) + k}{x^2}$.

Problem 2

Hint 1. By Lagrange, the two difference quotients are $f'(c_1)$ and $g'(c_2)$ for some $c_1, c_2 \in (a, b)$. Now study $h = f'g'$.

Hint 2. f' and g' are increasing and positive, so h is increasing, and if say $c_1 \leqslant c_2$ then $h(c_1) \leqslant f'(c_1)g'(c_2) \leqslant h(c_2)$.

Hint 3. A monotone derivative is continuous (monotone functions have only jump discontinuities, and Darboux forbids jumps), so h is continuous and the IVT on $[c_1, c_2]$ gives c.

Problem 3

Hint 1. Note that $\frac{\pi}{4} = \int_0^1 \frac{dx}{1+x^2}$, and compare $\frac{1}{1+x^2}$ with the two bounds.

Hint 2. For $0 < x < 1$: $\frac{1}{1+x} < \frac{1}{1+x^2} < \frac{1}{2x}$. Since $\int_0^1 (f(x) - \frac{1}{1+x^2})dx = 0$, the continuous function $f - \frac{1}{1+x^2}$ cannot have a constant strict sign on $(0, 1)$, so it vanishes at some $x_0 \in (0, 1)$.

Problem 4

Hint 1. Suppose that $f' \geqslant 1 + f^2$ everywhere on (a, b), and look at $\arctan f$.

Hint 2. Then $(\arctan f)' \geqslant 1$, so by Lagrange $\arctan f(b) - \arctan f(a) \geqslant b - a \geqslant \pi$. This is impossible, because arctan takes values in $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Problem 5

(a)

Hint 1. The zeros have an accumulation point $c \in [a, b]$, and $f(c) = 0$ by continuity.

Hint 2. If $c \in (a, b)$, compute $f'(c)$ as the limit of difference quotients along zeros $x_n \to c$: it is 0. This is forbidden, so $c \in \{a, b\}$.

(b)

Hint 1. Try $f(x) = x \sin \frac{1}{x}$ with $f(0) = 0$. At its zeros $x = \frac{1}{k\pi}$ compute $f'(x) = \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x} = \pm k\pi \neq 0$.

Problem 6

Hint 1. The right-hand side involves only the odd part of f. Consider $g(x) = f(x) - f(-x) - 2xf'(0) - cx^3$, where c is chosen so that $g(1) = 0$, i.e. $c = f(1) - f(-1) - 2f'(0)$.

Hint 2. g is odd, and $g(0) = g(1) = 0$, $g'(0) = 0$. Apply Rolle three times, using $g'(0) = 0$ and then $g''(0) = 0$ (as g'' is odd), to get $\eta \in (0,1)$ with $g'''(\eta) = 0$.

Hint 3. This gives $\frac{1}{2}(f'''(\eta) + f'''(-\eta)) = 3c$, i.e. the average of $\frac{f'''(\pm\eta)}{6}$ equals the right-hand side. f''' need not be continuous, so finish with Darboux' theorem for $f''' = (f'')'$ on $[-\eta, \eta]$.

Problem 7

Hint 1. By Lagrange the left-hand side equals $f'(c)$ for some c. Compare $f'(c)$ with $f(c) + \frac{f'(c)^2}{4f(c)}$.

Hint 2. AM–GM: $f + \frac{f'^2}{4f} \geqslant 2\sqrt{\frac{f'^2}{4}} = |f'| \geqslant f'$.

Problem 8

Note. As printed, the right-hand side is $\frac{f'(c_2)}{2c_1}$, which makes the claim trivial: take $c_1 = c_2 = \frac{a+b}{2}$. The intended statement is $\frac{f'(c_1)}{a+b} = \frac{f'(c_2)}{2c_2}$, and the hints follow it.

Hint 1. Write both sides as the same quantity built from $f(b) - f(a)$, using two different mean value theorems.

Hint 2. Lagrange: $f'(c_1) = \frac{f(b)-f(a)}{b-a}$. Cauchy with $g(x) = x^2$: $\frac{f'(c_2)}{2c_2} = \frac{f(b)-f(a)}{b^2-a^2}$. Divide the first by $a + b$.

Problem 9

Hint 1. Near 0, $f'' > 0$, so f' is increasing and negative. Note that f itself may be unbounded (e.g. $f = -\ln x$), so split $f(x) = \big(f(x) - f(y)\big) + f(y)$ with a fixed small y.

Hint 2. For $0 < x < y$, Lagrange gives $f(x) - f(y) = -f'(\xi)(y - x)$ with $f'(x) \leqslant f'(\xi) < 0$, hence $\left|\dfrac{f(x) - f(y)}{f'(x)}\right| \leqslant y - x < y$.

Hint 3. Also $\dfrac{f(y)}{f'(x)} \to 0$ as $x \to 0^+$. Hence $\limsup_{x\to 0^+} |f(x)/f'(x)| \leqslant y$ for every small y.

Problem 10

Hint 1. Associate the polynomial $P(x) = a_0 + a_1 x + \cdots + a_n x^n$ with the operator $P(D)$. The answer: exactly the sequences for which all roots of P are real.

Hint 2. For a real λ: $f' - \lambda f = e^{\lambda x}\big(e^{-\lambda x} f\big)'$. So if f has $m + 1$ zeros, Rolle gives m zeros of $f' - \lambda f$ between the outermost ones.

Hint 3. If all roots are real, factor $P(D) = a_n(D - \lambda_1)\cdots(D - \lambda_n)$ and apply Hint 2 n times.

Hint 4. If $P(\alpha \pm i\beta) = 0$ with $\beta \neq 0$, then $u(x) = e^{\alpha x} \sin \beta x$ satisfies $P(D)u = 0$ and has many simple sign changes. Let m be the multiplicity of 0 as a root of P (so $a_m \neq 0$), and take $f = u + \varepsilon x^m$ with small $\varepsilon \neq 0$. On a bounded interval, f still has $n + 1$ zeros, while $P(D)f = \varepsilon\, m!\, a_m \neq 0$ everywhere.